- CREATE DATABASE
- CREATE TABLE
- INSERT
- UPDATE
- DELETE
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- Aggregate functions (COUNT, SUM, AVG, MAX, MIN)

```
CREATE DATABASE PracticeDB;
USE PracticeDB;

CREATE TABLE employees (
    emp_id INT PRIMARY KEY AUTO_INCREMENT,
    emp_name VARCHAR(50),
    department VARCHAR(30),
    salary DECIMAL(10,2),
    age INT
);
```

Insert Records

```
INSERT INTO employees (emp_name, department, salary, age) VALUES
('Alice', 'HR', 4000, 29),
('Bob', 'IT', 6000, 32),
('Charlie', 'IT', 7000, 40),
('David', 'Finance', 5000, 28),
('Eve', 'Finance', 6500, 35),
```

('Frank', 'HR', 4500, 30),

('Grace', 'IT', 8000, 45),

('Helen', 'Finance', 5500, 38),

('Ian', 'Marketing', 4800, 27),

('Jane', 'Marketing', 5200, 31);

Update Records

Increase salary of IT employees by 10%

UPDATE employees

SET salary = salary * 1.10

WHERE department = 'IT';

Delete Records

Delete employees from Marketing earning less than 5000

DELETE FROM employees

WHERE department = 'Marketing' AND salary < 5000;

WHERE Clause

Show employees older than 30 years

SELECT * FROM employees

WHERE age > 30;

ORDER BY

Show employees ordered by salary (highest first)

SELECT * FROM employees

ORDER BY salary DESC;

Aggregate Functions

- **COUNT**

SELECT department, COUNT(*) AS total_employees FROM employees

GROUP BY department;

- **SUM**

SELECT department, SUM(salary) AS total_salary

FROM employees

GROUP BY department;

- **AVG**

SELECT department, AVG(salary) AS avg_salary

FROM employees

GROUP BY department;

- **MAX**

SELECT department, MAX(salary) AS highest_salary

FROM employees

GROUP BY department;

- **MIN**

SELECT department, MIN(salary) AS lowest_salary

FROM employees

GROUP BY department;

GROUP BY

Show average salary per department

SELECT department, AVG(salary) AS avg_salary

FROM employees

```
GROUP BY department;
```

HAVING

Show departments where average salary is greater than 6000

```
SELECT department, AVG(salary) AS avg_salary
FROM employees
GROUP BY department
HAVING AVG(salary) > 6000;
```

PRACTICE QUESTIONS (Try Yourself)

1. **Find employees with salary between 5000 and 7000.**
2. **List employees in IT department ordered by age.**
3. **Show the total number of employees in each department.**
4. **Find the department with the highest total salary.**
5. **Show the average salary of employees younger than 35.**
6. **Update Finance employees' salary by adding 500 bonus.**
7. **Delete employees from HR whose salary is less than 4200.**
8. **Show department names where the number of employees is more than 2.**
9. **Show max and min salary in Marketing.**
10. **Get employees whose name starts with "A".**

MySQL Practice

Q1. Find employees with salary between 5000 and 7000.

```
SELECT *
FROM employees
WHERE salary BETWEEN 5000 AND 7000;
```

Q2. List employees in IT department ordered by age.

```
SELECT emp_name, age, salary
FROM employees
WHERE department = 'IT'
ORDER BY age;
```

Q3. Show the total number of employees in each department.

```
SELECT department, COUNT(*) AS total_employees
FROM employees
GROUP BY department;
```

Q4. Find the department with the highest total salary.

```
SELECT department, SUM(salary) AS total_salary
FROM employees
GROUP BY department
ORDER BY total_salary DESC
LIMIT 1;
```

Q5. Show the average salary of employees younger than 35.

```
SELECT AVG(salary) AS avg_salary_under35
FROM employees
WHERE age < 35;
```

Q6. Update Finance employees' salary by adding 500 bonus.

```
UPDATE employees
SET salary = salary + 500
WHERE department = 'Finance';
```

Q7. Delete employees from HR whose salary is less than 4200.

```
DELETE FROM employees
```

```
WHERE department = 'HR' AND salary < 4200;
```

Q8. Show department names where the number of employees is more than 2.

```
SELECT department, COUNT(*) AS total_employees
FROM employees
GROUP BY department
HAVING COUNT(*) > 2;
```

Q9. Show max and min salary in Marketing.

```
SELECT MAX(salary) AS highest_salary,
       MIN(salary) AS lowest_salary
FROM employees
WHERE department = 'Marketing';
```

Q10. Get employees whose name starts with "A".

```
SELECT *
FROM employees
WHERE emp_name LIKE 'A%';
```

Quick Recap

INSERT / UPDATE / DELETE → modify data

- **WHERE / ORDER BY** → filtering & sorting
- **COUNT, SUM, AVG, MAX, MIN** → aggregation
- **GROUP BY** → grouping rows
- **HAVING** → filtering grouped results